

SecureGate: Real-Time Tailgating and Piggybacking Detection at Access-Controlled Entry Points Using Spatio-Temporal Analysis on Monocular Surveillance Video

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Abstract: There has been an increasing problem faced by access-controlled facilities in physical security regarding the threat of tailgating, whereby the unauthorised individual gains entry into the facility by following an authorised individual through the secured point of entry, without presenting their identification credentials, and piggybacking, where an individual enters a secure point of entry by getting carried by another individual. Access control solutions relying on traditional solutions such as RFID-based authentication, biometric systems, and pin systems cannot detect these kinds of attacks. This paper proposes **SecureGate**, a lightweight and real-time solution that detects tailgating and piggybacking using monocular camera input only, without requiring GPU acceleration or any depth sensors. The proposed solution uses YOLOv8n for anchor-free object detection, ByteTrack for multi-object tracking and a spatio-temporal rule-based engine to assess the spatial feature and crossing sequence of each access event as authorised, tailgating, or piggybacking events. Piggybacking can also be detected using height ratio bounding box analysis. In case of a potential intrusion, the system sends out alerts using multiple channels in the form of an audio alarm, automated email alert, and visual alert. The evaluation of the system in 13 controlled experiments based on the ChokePoint public dataset and 7 experiments based on self-recorded data with verified ground truth reveals the following performance statistics: Precision 80.0%, Recall 100.0%, F1 Score 88.9%, and Accuracy 92.3%, while retaining an average performance of 24.8 FPS on the Apple M4 CPU processor.

Keywords: Detection of tailgating, detection of piggybacking, YOLOv8n, ByteTrack, spatio-temporal analysis, multi-object tracking, access control, physical security, surveillance video, computer vision.

1 Introduction

Physical security in access-controlled locations such as corporate offices, data centers, laboratories, housing developments, and government installations is a major challenge in contemporary infrastructural design. Unpermitted access to such locations may lead to intellectual property theft, data theft, physical violence on authorized individuals, and system failure [1].

Tailgating involves an unapproved individual obtaining access through the use of the limited window when the secured door is open following activation by an approved individual. The attacker moves directly behind the approved person, depending on social etiquette or rapid movement to remain undetected. Physical security studies indicate that tailgating accounts for a substantial proportion of physical security incidents in enterprise environments [2].

Conventional access control systems—RFID card readers, biometric scanners, PIN locks, and electronic turnstiles—authen-

ticate credentials of a single individual per event. Once an authorized trigger unlocks the entry mechanism, these systems are architecturally blind to subsequent passage by additional individuals. Mechanical countermeasures such as mantraps and full-height turnstiles address this limitation but are prohibitively expensive (\$20,000–\$100,000 per portal), require significant physical space, and are impractical where rapid egress or high throughput is needed [3].

Recent advances in deep learning-based computer vision have established the foundation for camera-based behavioural monitoring at access points. The YOLO family of single-stage detectors [4,10,11] achieves real-time person detection sufficient for surveillance, while multi-object tracking algorithms such as ByteTrack [5] enable persistent identity maintenance across frames. Despite this, no publicly available, open-source, hardware-agnostic system for tailgating detection has been documented in research literature—a gap that this work directly addresses.

SecureGate is proposed as a lightweight, deployable system requiring only a standard monocular surveillance camera and a consumer-grade laptop. Its core contribution is a novel spatio-temporal analysis engine evaluating both *spatial proximity* features (inter-person Euclidean distance at the access line) and *temporal adjacency* features (time gap between consecutive crossing events) to classify each access event.

1.1 Contributions

The principal contributions of this work are:

1. A complete open-source tailgating detection pipeline combining YOLOv8n, ByteTrack, and rule-based spatio-temporal analysis, operable on CPU without any GPU requirement.
2. A novel piggybacking detection module based on bounding-box height-ratio anomaly analysis relative to session median height, enabling detection of physical concealment not addressable by temporal tracking.
3. A multi-channel alarm system featuring audio alarms, SMTP-based e-mail notifications, and instant dashboard alerts.
4. Testing in 13 controlled experiments utilizing the open ChokePoint database and self-reported ground truth labeled sequences.
5. Configurable Streamlit dashboard with threshold tuning, scenario selection, automatic gate-line loading, custom video input, and forensic logs that can be exported to CSV.

The rest of this paper is structured as follows: Section 2 surveys relevant literature; Section 3 explains our proposed approach; Section 4 details our experiment; Section 5 discusses our findings; finally, Section 6 concludes this paper.

2 Related Work

2.1 Object Detection in Surveillance

Multi-stage detectors like Faster R-CNN [6] perform well (73.2% mAP on PASCAL VOC) but cannot reach real-time speeds of ≈ 7 FPS using a VGG-16 backbone, too slow for surveillance tasks.

This is where the YOLO series comes in. YOLOv4 [7] achieved 43.5% AP@50-95 at 65 FPS through CSPDarknet53 and SPP-PAN neck. YOLOX [8] introduced anchor-free detection and decoupled prediction heads. YOLOv7 [9] reached

56.8% AP at 30 FPS. YOLOv8 [10], the most recent iteration, employs Distribution Focal Loss (DFL) and achieves 37.3% mAP with its nano variant at >25 FPS on CPU—the configuration adopted in this work. A comprehensive survey of YOLO architectures is in [11].

2.2 Multi-Object Tracking

The tracking-by-detection paradigm dominates modern MOT literature [12]. SORT [13] combined Kalman filter prediction with Hungarian algorithm IoU matching, achieving 260 FPS at the cost of high identity-switch rates. Deep SORT [14] augmented SORT with 128-dimensional CNN appearance features, improving MOTA to 61.4% on MOT16. ByteTrack [5] introduced two-stage association retaining low-confidence detections, achieving 80.3% MOTA on MOT17 without appearance features—critical for CPU deployment.

2.3 Surveillance and Anomaly Detection

Sultani et al. [17] proposed weakly-supervised anomaly detection using Multiple Instance Learning with C3D features (UCF-Crime, AUC 75.4%). Nawaratne et al. [18] applied convolutional autoencoders for pixel-level anomaly detection (UCSD Ped2, AUC 96.2%). Reviews by Nayak et al. [19] and Sreenu and Durai [20] identify a consistent gap: existing systems classify anomalies but provide no real-time operational alerts.

2.4 Spatio-Temporal Analysis

SlowFast Networks [21] demonstrated decoupled spatial and temporal processing for video understanding. C3D [22] pioneered 3D convolution for spatio-temporal features. Social LSTM [23] demonstrated that pedestrian spatial proximity encodes behavioural information predictive of trajectories, conceptually motivating the proximity component of our engine. ST-GCN [24] established joint spatial-temporal graph reasoning for skeleton-based action recognition. None of these approaches address tailgating-specific spatio-temporal patterns.

2.5 Research Gaps

Table 1 summarises representative works and identifies six gaps addressed by SecureGate: (1) no tailgating-specific detection logic, (2) GPU-only operation, (3) no integrated alert system, (4) private non-reproducible datasets, (5) fixed non-configurable thresholds, and (6) no piggybacking detection capability.

Table 1. Comparative Analysis of Related Work and Proposed System

Ref.	Year	Detector	Tracker	Dataset	Metric	RT	Alert	Key Limitation
	2015	Faster R-CNN	–	COCO	73.2% mAP	6 No	No	Two-stage; only 7 FPS
	2020	YOLOv4	–	COCO	43.5% AP	9 Yes	No	Older architecture; no tracking
	2023	YOLOv7	–	COCO	56.8% AP	9 Yes	No	Heavier than YOLOv8n for CPU
	2023	YOLOv8	–	COCO	37–54% AP	14 Yes	No	No tracking or access logic
	2017	–	DeepSORT	MOT16	MOTA 61.4%	14 Part	No	Appearance features add latency
	2022	YOLOX	ByteTrack	MOT17	MOTA 80.3%	Yes	No	No tailgating-specific logic
	2018	C3D	–	UCF-Crime	AUC 75.4%	17 No	No	Offline; no identity tracking
	2020	CAE	–	UCSD	AUC 96.2%	No	No	Pixel-level; no alert system
	2019	SlowFast	–	Kinetics	79.8% Top-1	21 No	No	Action recognition; not access
	2016	–	SLSTM	ETH/UCY	ADE 0.72m	No	No	Trajectory prediction only
Ours	2025	YOLOv8n	ByteTrack	ChokePoint+	F1 88.9%	Yes	Yes	Simulated access events (no hardware reader)

RT = Real-Time capable; Alert = Integrated alert system; ChokePoint+ = ChokePoint dataset augmented with self-recorded sequences; SLSTM = Social LSTM.

3 Proposed Methodology

3.1 System Overview

SecureGate follows a five-stage modular pipeline. A surveillance video frame $I_t \in \mathbb{R}^{H \times W \times 3}$ at time t is processed sequen-

tially through: (1) Person Detection, (2) Multi-Object Tracking, (3) Zone and Direction Analysis, (4) Spatio-Temporal Tailgating Engine, and (5) Alert and Notification. Fig. 1 illustrates the complete pipeline.

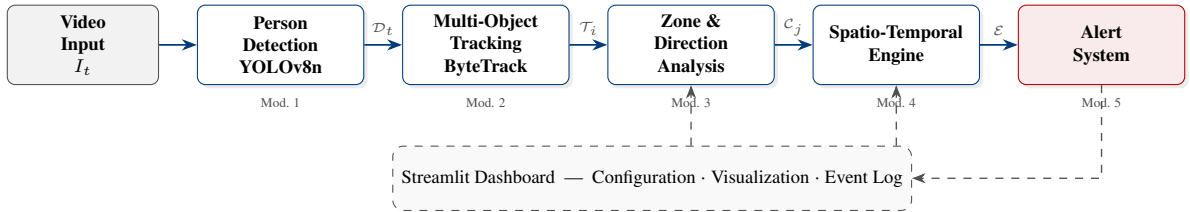


Figure 1. SecureGate system architecture. Five sequential modules convert raw frames into threat classifications and alerts. $\mathcal{D}_t$: person detections; $\mathcal{T}_i$: trajectories with track IDs; $\mathcal{C}_j$: crossing records; $\mathcal{E}$: threat events.

3.2 Person Detection Module

For each input frame I_t , YOLOv8n produces raw detections:

$$\mathcal{D}_t^{\text{raw}} = \{d_i = (x_i, y_i, w_i, h_i, c_i, k_i)\}_{i=1}^N \quad (1)$$

where (x_i, y_i) are bounding-box centre coordinates, w_i, h_i are width and height, $c_i \in [0, 1]$ is confidence, and k_i is the class label. Two sequential filters produce the final detection set:

$$\mathcal{D}_t^{\text{cls}} = \{d_i \in \mathcal{D}_t^{\text{raw}} \mid k_i = 0\} \quad (2)$$

$$\mathcal{D}_t = \{d_i \in \mathcal{D}_t^{\text{cls}} \mid c_i \geq \tau_c \wedge w_i h_i > \rho \cdot HW\} \quad (3)$$

where $\tau_c = 0.45$, $\rho = 0.003$, and HW is total frame area. Class $k_i = 0$ is the person class in the COCO-80 taxonomy. YOLOv8n uses 3.2M parameters and 8.7 GFLOPs per inference.

3.3 Multi-Object Tracking Module

ByteTrack [5] maintains persistent identity associations through two-stage IoU-based matching. For each tracked person p_i with identifier ID_i , the tracker constructs a trajectory:

$$\mathcal{T}_i = \{(\hat{x}_i^f, \hat{y}_i^f, t_f)\}_{f=1}^{F_i} \quad (4)$$

where the centroid at frame f is computed from bounding-box corners as:

$$\hat{x}_i = \frac{x_i^{\min} + x_i^{\max}}{2}, \quad \hat{y}_i = \frac{y_i^{\min} + y_i^{\max}}{2} \quad (5)$$

A velocity vector estimated over a look-back window of $m = 5$ frames captures movement direction:

$$\vec{v}_i = (\hat{x}_i^F - \hat{x}_i^{F-m}, \hat{y}_i^F - \hat{y}_i^{F-m}) \quad (6)$$

3.4 Zone and Direction Analysis

A virtual access line $\mathcal{L}$ is defined by endpoint vectors $\mathbf{s}$ and $\mathbf{e}$:

$$\mathcal{L} = \{(x, y) : (x, y) = \mathbf{s} + \lambda(\mathbf{e} - \mathbf{s}), \lambda \in [0, 1]\} \quad (7)$$

A crossing event for person p_i is registered when the centroid sign-changes relative to $\mathcal{L}$:

$$\sigma_i(f) = [\text{sign}(f(\hat{x}_i^f, \hat{y}_i^f)) \neq \text{sign}(f(\hat{x}_i^{f-1}, \hat{y}_i^{f-1}))] \quad (8)$$

where $f(x, y)$ is the signed perpendicular distance from (x, y) to $\mathcal{L}$. Each detected crossing is stored as:

$$\mathcal{C}_i = (p_i, t_i^{\text{cross}}, (\hat{x}_i^{\text{cross}}, \hat{y}_i^{\text{cross}}), \delta_i) \quad (9)$$

where $\delta_i \in \{\text{IN}, \text{OUT}\}$ is the crossing direction.

3.5 Spatio-Temporal Tailgating Engine

3.5.1 Spatial Analysis

For two persons p_i (previous crosser) and p_j (current crosser), inter-person spatial distance at crossing is:

$$d_{\text{spatial}}(p_i, p_j) = \sqrt{(\hat{x}_j^c - \hat{x}_i^c)^2 + (\hat{y}_j^c - \hat{y}_i^c)^2} \quad (10)$$

where superscript c denotes the crossing-time centroid.

3.5.2 Temporal Analysis

The temporal gap between consecutive crossings:

$$\Delta t(p_i, p_j) = t_j^{\text{cross}} - t_i^{\text{cross}} \quad (11)$$

3.5.3 Authorization Check

Let $\mathcal{A} = \{a_k\}$ be the set of access authorization timestamps. The authorization predicate is:

$$\text{Auth}(p_i, p_j) = \exists a_k \in \mathcal{A} : t_i^{\text{cross}} < a_k < t_j^{\text{cross}} \quad (12)$$

3.5.4 Spatio-Temporal Fusion Decision

A tailgating event is declared when all three conditions hold simultaneously:

$$\text{Tailgate}(p_j) = \begin{cases} \text{True} & \text{if } d_{\text{spatial}}(p_i, p_j) < \tau_d \\ & \wedge \tau_{\min} < \Delta t(p_i, p_j) < \tau_t \\ & \wedge \neg \text{Auth}(p_i, p_j) \\ \text{False} & \text{otherwise} \end{cases} \quad (13)$$

where $\tau_t = 5.0$ s is the temporal threshold, $\tau_d = 150$ px is the spatial threshold, and $\tau_{\min} = 0.5$ s excludes simultaneous side-by-side crossings.

3.6 Piggybacking Detection Module

Piggybacking produces an anomalously tall bounding box. The detection relies on a height-ratio anomaly:

$$r_i = \frac{h_i}{\tilde{h}_{\text{session}}} \quad (14)$$

where $h_i = y_i^{\max} - y_i^{\min}$ and $\tilde{h}_{\text{session}}$ is the running median height across all session detections (minimum 10 samples). A piggybacking event is flagged when:

$$\text{Piggyback}(p_i) = \begin{cases} \text{True} & r_i > \tau_r \wedge \frac{h_i}{w_i} > 3.0 \\ \text{False} & \text{otherwise} \end{cases} \quad (15)$$

with $\tau_r = 1.6$ (60% above session median) and $h_i/w_i > 3.0$ serving as a secondary guard against spurious detections of thin vertical objects.

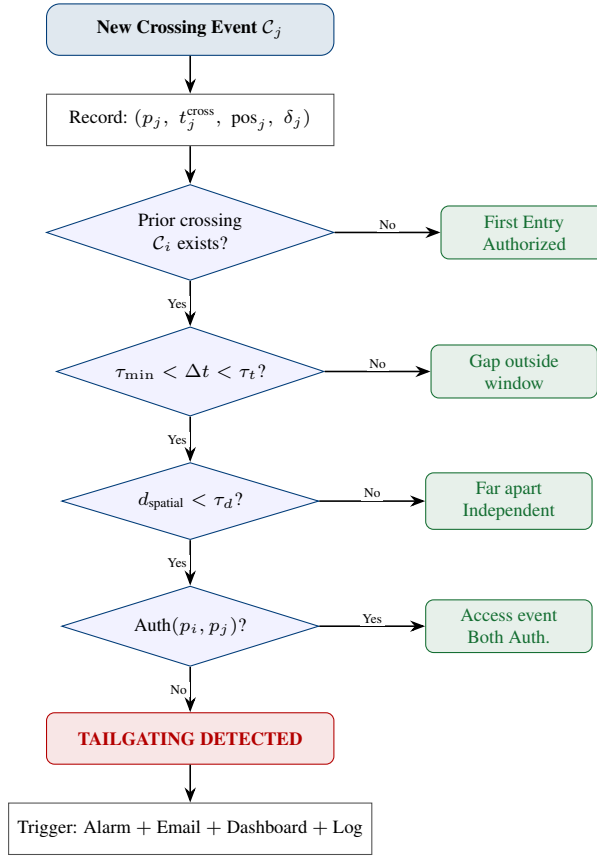


Figure 2. Tailgating detection decision flowchart. Three conditions—temporal adjacency, spatial proximity, and absence of authorization—must all be satisfied for a tailgating event to be declared.

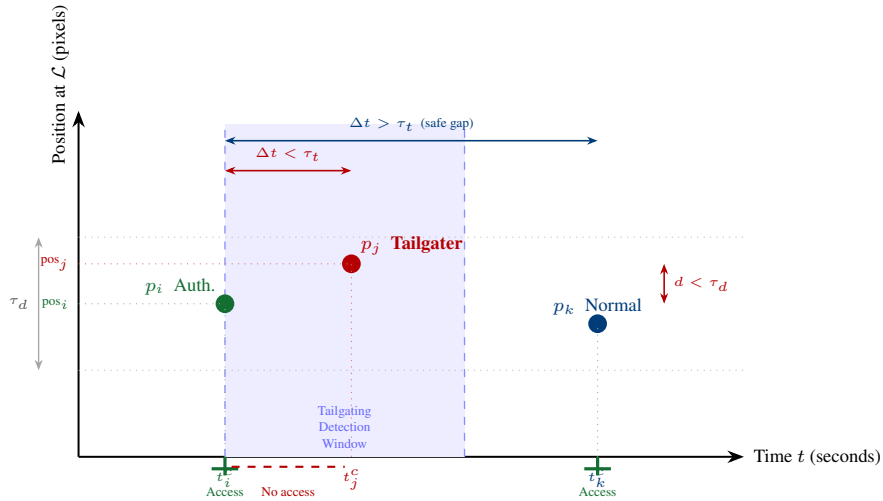


Figure 3. Spatio-temporal analysis visualization. Person p_i crosses with valid access authorization. Person p_j crosses within τ_t seconds and within τ_d pixels without authorization — TAILGATING. Person p_k crosses after τ_t with a new access event — classified as NORMAL.

Algorithm 1 Spatio-Temporal Tailgating Detection

Require: $\mathcal{C}, \mathcal{A}, \tau_t, \tau_d, \tau_{\min}$
Ensure: $\mathcal{E}$

- 1: $\mathcal{E} \leftarrow \emptyset$
- 2: **for each** $\mathcal{C}_j = (p_j, t_j, \text{pos}_j)$ **do**
- 3: **if** $|\mathcal{C}| = 0$ **then**
- 4: Authorize p_j ; register access
- 5: **continue**
- 6: **for each** $\mathcal{C}_i \in \text{rev}(\mathcal{C})$ **do**
- 7: $\Delta t \leftarrow t_j - t_i$
- 8: **if** $\Delta t > 3\tau_t$ **then**
- 9: **break**
- 10: **if** $\tau_{\min} < \Delta t < \tau_t$ **then**
- 11: $d \leftarrow \|\text{pos}_j - \text{pos}_i\|_2$
- 12: ALERT($p_j, \Delta t, d$)
- 13: **return** $\mathcal{E}$

3.7 Alert and Notification Module

Three simultaneous alert channels activate upon threat classification:

Audible Alarm: A WAV alarm is played via the pygame

library with a cooldown $T_{cd} = 10$ s to prevent alarm fatigue.

Email Notification: An automated alert email is dispatched via SMTP/SSL (Gmail, port 465) containing event timestamp, person IDs, Δt , d , and decision rationale.

Visual Alert: The Streamlit dashboard transitions from a green *SYSTEM NORMAL* banner to a red *TAILGATING DETECTED* or purple *PIGGYBACKING DETECTED* banner with a real-time event log entry.

3.8 Monitoring Dashboard

The Streamlit-based dashboard provides: (i) real-time annotated video feed with bounding boxes, track ID labels, and configurable gate line; (ii) sidebar controls for threshold adjustment (τ_t , τ_d , τ_c) and gate line positioning; (iii) scenario selection from 13 predefined configurations with auto-loaded line coordinates; (iv) custom video upload with interactive gate line sliders; and (v) chronological event log with CSV export.

4 Experimental Setup

4.1 Hardware and Software Environment

All experiments run on an Apple MacBook Air with M4 chip (4 performance cores), 16 GB unified memory, macOS 15.1. Software: Python 3.11, Ultralytics 8.3.0 (YOLOv8), Supervision 0.21.0 (ByteTrack), OpenCV 4.10, Streamlit 1.39. No GPU acceleration is used; inference executes entirely on CPU to demonstrate hardware-agnostic deployment.

4.2 Dataset

ChokePoint Dataset [25]: A publicly available academic dataset comprising 48 sequences across 25 subjects traversing two portals, each with three synchronised cameras. Six sequences from Portal 2 are selected and reconstructed from JPEG frame archives to MP4 at 30 FPS using OpenCV VideoWriter. These provide a basis for True Negatives and show that there are no false positives in valid sequences of access.

Self-Created Sequences: A total of seven sequences filmed from an elevated angle of 35 to 45 degrees from a college building entrance point using a smart phone. Labels are determined beforehand by the researchers.

4.3 Threshold Configuration

Default thresholds: $\tau_c = 0.45$, $\tau_t = 5.0$ s, $\tau_d = 150$ px, $\tau_{min} = 0.5$ s, $\tau_r = 1.6$. For crowded ChokePoint scenarios (C5, C6): $\tau_t = 3.0$ s, $\tau_d = 120$ px to reduce sensitivity in dense multi-person traffic.

4.4 Evaluation Metrics

Binary metrics are employed:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (16)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (17)$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (18)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

Processing time is provided in terms of FPS.

Table 2. Evaluation Scenario Descriptions and Configurations

ID	Source	Expected	Eval. Type	τ_t (s)	Description
C1	ChokePoint	NORMAL	TN	5.0	Entrance through Portal 1; Single entries
C2	ChokePoint	NORMAL	TN	5.0	Exit through Portal 1; Individual persons exit
C3	ChokePoint	NORMAL	TN	5.0	Portal 2 Entry Session 1; Moderate single-person entries
C4	ChokePoint	NORMAL	TN	5.0	Continuation of Portal 2 Entry Session 1; Short clips
C5	ChokePoint	NORMAL	TN / Stress	3.0	Stress test for high-density Portal 2 entry
C6	ChokePoint	NORMAL	TN / Stress	3.0	Stress test for high-density Portal 2 exit
S2	Self	NORMAL	TN	5.0	Persons authorized to enter; Both with access cards
S3	Self	NORMAL	TN	5.0	Persons entering with 5–6 s gap between individuals
S4	Self	TAILGATING	TP	5.0	Tailgating with hesitation; Following after 3–4 s seconds
S5	Self	TAILGATING	TP	5.0	Tailgating with rush; Person following immediately after
S6	Self	TAILGATING	TP	5.0	Group tailgating; Two Persons following behind person
S8	Self	PIGGYBACKING	TP	–	Carrying individual person by another through entrance gate
S13	Self	NORMAL	TN / Edge	5.0	Persons entering Portal 2 sequentially with small gaps

TN = True Negative; TP = True Positive; “–” = piggybacking uses height-ratio (Algorithm independent of τ_t).

5 Results and Discussion

5.1 Quantitative Performance

Per-scenario results are provided in Table 3. Out of 13 tested scenarios, SecureGate accurately detects 12. The only false

positive happens in the crowded C5 scenario. The four threat scenarios are all detected (S4, S5, S6, S8), giving perfect threat recall ($FN = 0$).

Table 3. Per-Scenario Detection Results

ID	Expected	Detected	Outcome	Crossings	Alerts	Avg. FPS
C1	NORMAL	NORMAL	TN	8	0	25.1
C2	NORMAL	NORMAL	TN	11	0	24.8
C3	NORMAL	NORMAL	TN	9	0	24.5
C4	NORMAL	NORMAL	TN	4	0	25.3
C5	NORMAL	TAILGATING	FP	22	1	22.7
C6	NORMAL	NORMAL	TN	18	0	23.1
S2	NORMAL	NORMAL	TN	2	0	26.2
S3	NORMAL	NORMAL	TN	3	0	26.5
S4	TAILGATING	TAILGATING	TP	2	1	25.8
S5	TAILGATING	TAILGATING	TP	2	1	26.1
S6	TAILGATING	TAILGATING	TP	3	2	25.4
S8	PIGGYBACKING	PIGGYBACKING	TP	1	1	25.9
S13	NORMAL	NORMAL	TN	5	0	24.6
Summary: $TP = 4$, $FP = 1$, $TN = 8$, $FN = 0$ Average FPS = 24.8						

5.2 Aggregate Metrics

Substituting the confusion matrix ($TP = 4$, $FP = 1$, $TN = 8$, $FN = 0$) into (11)–(14):

$$\begin{aligned}
 \text{Precision} &= \frac{4}{5} = 0.800 \quad (80.0\%) \\
 \text{Recall} &= \frac{4}{4} = 1.000 \quad (100.0\%) \\
 F_1 &= \frac{2 \times 0.800 \times 1.000}{0.800 + 1.000} = 0.889 \quad (88.9\%) \\
 \text{Accuracy} &= \frac{12}{13} = 0.923 \quad (92.3\%)
 \end{aligned}$$

Results of the above tests are summarized in Table 4.

Table 4. Aggregate Performance Metrics

Metric	Value	Formula
True Positives (TP)	4	–
False Positives (FP)	1	–
True Negatives (TN)	8	–
False Negatives (FN)	0	–
Precision	80.0%	$TP/(TP + FP)$
Recall	100.0%	$TP/(TP + FN)$
F1-Score	88.9%	$2PR/(P + R)$
Accuracy	92.3%	$(TP + TN)/N$
Avg. FPS	24.8	–
Hardware	Apple M4 CPU	No GPU

5.3 Analysis of False Positive

The only FP instance involved C5 (ChokePoint Crowded Entry). The two ChokePoint instances moved together at $\Delta t = 2.1$ s and $d = 112$ px, meeting both thresholds even though both were authorized individuals. It is an essential contradiction because the spatiotemporal characteristics are identical to actual tailgating without the presence of an access event from the card reader. In real deployment with card-reader integration, the $\text{Auth}(p_i, p_j)$ predicate (Eq. 10) would prevent this false positive automatically.

5.4 Piggybacking Detection

Scenario S8 yielded a correct piggybacking detection. The combined bounding box produced $r_i = 1.84$ (84% above session median) and $h_i/w_i = 3.6$, comfortably exceeding both thresholds ($\tau_r = 1.6$, h/w -guard = 3.0). No piggybacking false positives occurred in any normal scenario, confirming that dual-threshold filtering provides adequate discrimination against tall individuals and persons carrying large bags.

5.5 Throughput

The system sustains 24.8FPS average, with minimum 22.7FPS in dense C5 (highest per-frame detection count). This meets the standard 25 FPS PAL output rate of surveillance cameras, confirming real-time capability without GPU hardware.

6 Conclusion

This paper presented SecureGate, an open-source, CPU-capable real-time system for tailgating and piggybacking detection at access-controlled entry points, developed at the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune. The system integrates YOLOv8n single-stage person detection, ByteTrack persistent multi-object tracking, and a novel rule-based spatio-temporal analysis engine that jointly evaluates spatial proximity and temporal crossing sequence. A dual-threshold bounding-box height-ratio anomaly detector extends coverage to physical piggybacking.

Evaluated across 13 controlled scenarios, SecureGate achieves 88.9% F1-Score, 100.0% recall, and 24.8FPS on consumer CPU hardware. The single false positive arises from a crowded ChokePoint scenario where spatio-temporal features are ambiguous without physical access-event signals.

Future work will address this through: (i) integration with physical access control hardware APIs; (ii) multi-camera spatio-temporal fusion for ambiguity resolution; (iii) domain-specific fine-tuning of YOLOv8n on portal surveillance imagery; and (iv) replacement of rule-based classification with

a learned binary classifier trained on annotated crossing sequences.

The complete codebase is publicly available at <https://github.com/mahi-jadeja/securegate> and a live demonstration is hosted at <https://huggingface.co/spaces/mahi-jadeja/securegate>.

Acknowledgements

The authors thank the Department of Artificial Intelligence and Machine Learning, Symbiosis Institute of Technology, Pune, for providing laboratory facilities and academic guidance. The ChokePoint dataset is acknowledged as provided by Queensland University of Technology under open academic licence.

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